



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No.						
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Institute: (0203)SYMBIOSIS CENTRE FOR MANAGEMENT AND HUMAN RESOURCE DEVELOPMENT,
PUNE

Programme: (020341) MASTER OF BUSINESS ADMINISTRATION

Batch: 2023-25,2024-26

Semester: I

Course: Operations Research

Course Code: 0203410107

Date: 15/03/2025

Maximum Marks: 40

Day: Saturday

Time: 14:00 - 15:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- 1) All questions are compulsory.
- 2) Use of calculators is allowed.

Q.1 A company makes two types of sofas, regular and long, at two locations, one in Hickory and one in Lenoir. The plant in Hickory has a daily operating budget of \$45,000 and can produce at most 300 sofas daily in any combination. It costs \$150 to make a regular sofa and \$200 to make a long sofa at the Hickory plant. The Lenoir plant has a daily operating budget of \$36,000, can produce at most 250 sofas daily in any combination and makes a regular sofa for \$135 and a long sofa for \$180. The company wants to limit production to a maximum of 250 regular sofas and 350 long sofas each day. If the company makes a profit of \$50 on each regular sofa and \$70 on each long sofa, how many of each type should be made at each plant in order to maximize profit? Formulate as an LPP. **(CO1) (7 marks)**

Q.2 The coach of a junior Quadrathlon needs to assign athletes to a medley relay team to send to junior Olympics. Since most of his best athletes are good in more than one sport it is not clear which athlete should be assigned to which sport. The five athletes and their best times (in seconds) are given below. Make an optimum assignment so that the best team is selected. Also state who will miss the opportunity of being part of the team. **(CO2) (6 marks)**

Stroke	Ravi	Aaryan	Ritwik	Murugan	Tony
Swimming	37.5	33.0	33.8	36.7	35.5
Kayaking	43.5	32.8	41.6	35.1	43.0
Cycling	33.5	28.8	39.0	30.7	34.1
Running	29.6	26.8	28.5	28.3	30.8

Q.3 The following transportation problem represents cost per unit of transportation from five plants to four warehouses. Find the optimal allocations and the least transportation cost. (CO3) (12 marks)

	WH1	WH2	WH3	WH4	Supply
Plant1	6	6	11	15	80
Plant2	4	6	10	12	120
Plant3	6	4	7	6	150
Plant4	4	10	14	14	70
Plant5	8	8	7	9	90
Demand	100	200	120	80	

Q.4 The following table provides the sensitivity report of an LP problem.

Adjustable Cells

Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
x1	0	-0.0604	0.026	0.0604	1E+30
x2	0	-0.0355	0.0509	0.0355	1E+30
x3	8.4	0	0.0864	2083018978	0.0084
x4	0	-0.00925	0.06875	0.00925	1E+30
x5	5.6	0	0.078	0.0084	0.00925

Constraints

Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
	2.78801E-				
Farm and comm	10	0.0084	0	5.6	8.4
Home	-4.2	0	0	1E+30	4.2
Bad debts	-0.196	0	0	1E+30	0.196
Total	14	0.08304	14	1E+30	14

Analyse the sensitivity report and answer the following questions.

(CO3) (10 marks)

1. What is the value of the objective function at optimal solution?
2. Does the problem represent a minimizing objective function or a maximizing function? Why?
3. Which constraints are binding constraints and why?
4. How much should be the objective coefficient of X2 if the variable X2 is to be included as a part of the optimal solution?
5. If the 'Total' constraint is reduced to 10, what will be the resultant change in the objective function value? Also, state what kind (less than or more than) of constraint it is.

Q.5 The following table shows the payoffs in \$ million for three strategies in two states of nature. Find best alternative using EMV. Also calculate the expected value of perfect information.

(CO2) (5 marks)

	Positive growth (0.65)	Slowdown (0.15)	No change (0.2)
Strategy 1	77	45	52
Strategy 2	121	38	87
Strategy 3	157	- 43	69
Strategy 4	112	-18	95